

8	(a)	The cut-off wavelength corresponds to the most energetic photon that can be produced. That happens <u>when all the kinetic energy of an accelerated electron is lost in a single collision/interaction with the target atom in producing one photon.</u>	A1
	(b)	When electrons striking the metal target interact with the crystal lattice, forces experienced by the electrons cause them to be accelerated, decelerated or deflected. When this occurs, <u>their kinetic energies are lost through the emission of <i>Bremsstrahlung</i> (or "braking radiation"), which are photons of a range of energies which can lie in the X-ray region.</u> Since <u>the magnitude of the 'deceleration' experienced by the incident electrons is different for all and is not discrete</u>, the wavelengths of the emitted photons have a continuous distribution so the <i>Bremsstrahlung</i> produces a continuous spectrum of electromagnetic radiation. OR <i>Bremsstrahlung</i> radiation, which is emitted when high energy external electrons coming close to the nucleus decelerate, accelerate or deflect. This energy lost in terms of photons can be any amount of energy less than the maximum kinetic energy of the electrons, therefore forming continuous spectra.	A1 A1

